



RESPONSIBLE AI DASHBOARD FOR MODEL RISK ASSESSMENT AND SUSTAINABILITY MONITORING

AI, Power & Responsibility – Governing Intelligent Systems for Trustworthy AI



Transparency



Fairness



Accountability



Governance

01 MOTIVATION

Artificial Intelligence is increasingly deployed in cybersecurity, sustainability, healthcare and financial systems. However, explainability, fairness, stability monitoring, drift detection and governance challenges reduce trust in AI-driven decisions.



Goal

Develop a Responsible AI Dashboard integrating explainability, fairness, stability monitoring, drift detection, governance risk assessment and sustainability monitoring.

DATASETS



Cybersecurity Intrusion Dataset

- Network traffic records
- Authentication logs
- Attack labels

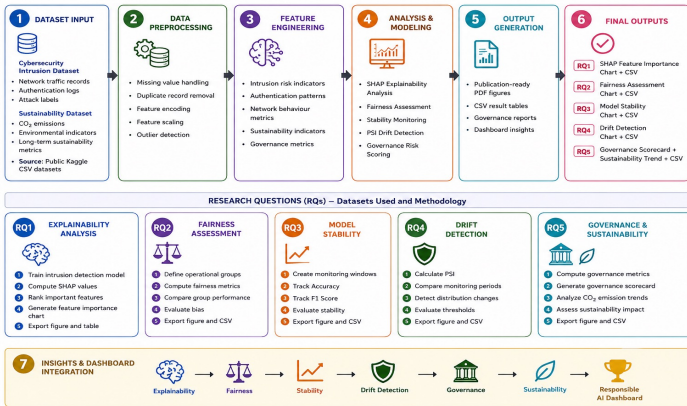


Sustainability / Environmental Dataset

- Sustainability & Environmental Indicators
- CO₂ Emissions
- Long term sustainability metrics

All datasets are publicly available for research purposes.

METHODOLOGY WORKFLOW



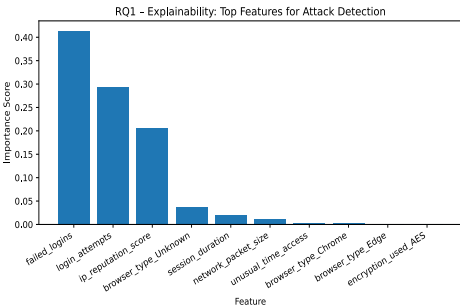
? RESEARCH QUESTIONS

- RQ1** How explainable are AI model predictions?
- RQ2** Does Fairness assessment indicated low bias across network activity groups?
- RQ3** How can governance and sustainability indicators improve Responsible AI oversight?
- RQ4** Can model drift be detected automatically?
- RQ5** How can a dashboard improve AI governance and accountability?

KEY RESULTS (HIGHLIGHTS)

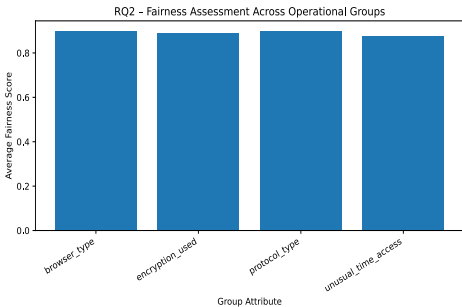
- SHAP analysis identified the most influential features driving intrusion detection predictions.
- Fairness assessment indicated low bias across network activity groups.
- The model achieved **88.47% accuracy** while maintaining stable prediction performance.
- Drift detection reported a minimal drift score of **0.0043**, indicating stable operational behavior.
- The Responsible AI Dashboard generated an **Overall Risk Score of 86/100**, supporting governance and compliance decisions.

RQ1 – Feature Importance (SHAP)



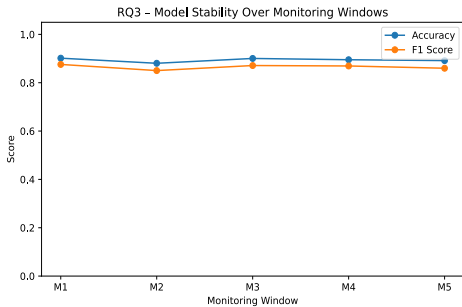
SHAP values show the impact of each feature on model predictions.

RQ2 – Fairness Assessment



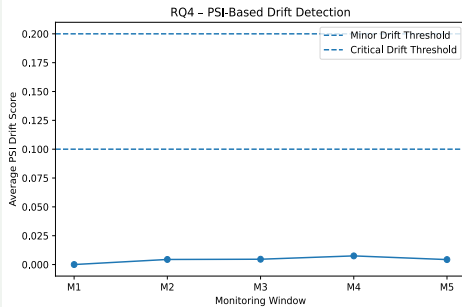
Fairness scores closer to 1 indicate lower bias across demographic groups.

RQ3 – Model Performance Over Time



Slight performance degradation observed over time.

RQ4 – Drift Detection



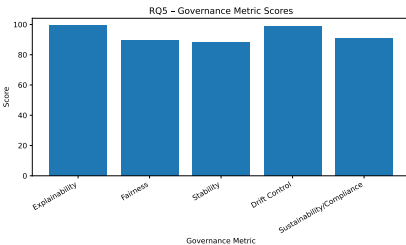
Drift score increases over time indicating distribution changes in the data.

RQ5 – Governance & Sustainability Assessment

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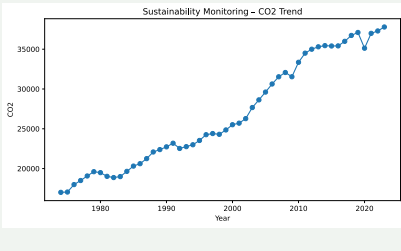
Overall Risk Score
(out of 100)

- 0.00 – 0.05: No Drift
- 0.05 – 0.15: Minor Drift
- >0.15: Critical Drift



Aggregated risk score to support governance decisions.

Metric	Score
Explainability Score	99/100
Fairness Score	89/100
Stability Score	88/100
Compliance Score	90/100
Overall Risk Score	86/100



RESPONSIBLE DASHBOARD RESULTS

Model Accuracy 88.47% ↑ 2% vs last month	Fairness Score 89.42% ↑ 3% vs last month	Drift Score 0.0043 Stable	Explainability Score 99.14% ↑ 5% vs last month	Compliance Status PASS All checks passed	Overall Risk Score 86/100 Low Risk
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Alerts & Insights

- Drift score remains within acceptable threshold.
- Explainability analysis identified high-impact prediction features.
- Fairness metrics remain within governance limits.
- Continue periodic monitoring and model validation.

CONTRIBUTIONS

- Comprehensive Responsible AI Dashboard for model monitoring.
- Integration of Explainability (SHAP) for transparency.
- Bias and Fairness assessment across operational network groups.
- Automated Drift Detection for continuous monitoring.
- Risk Scoring Engine for actionable insights.
- Supports Governance, Compliance and Accountability.

08 CONCLUSION

- The Responsible AI Dashboard provides a comprehensive framework for monitoring, evaluating, and governing AI systems throughout their lifecycle. By integrating explainability, fairness assessment, performance tracking, drift detection, and compliance monitoring, the dashboard enables organizations to identify risks before they impact decision-making processes.
- The results demonstrate that continuous monitoring significantly improves transparency, accountability, and trustworthiness in AI deployments. The unified risk scoring mechanism simplifies complex model evaluation metrics into actionable insights for stakeholders, regulators, and decision-makers.
- Overall, this work highlights the importance of Responsible AI practices and demonstrates how governance-driven dashboards can support ethical, fair, and reliable AI systems in real-world applications.
- The proposed framework achieved an **Explainability Score of 99.14%**, **Fairness Score of 89.42%**, and **Overall Risk Score of 86/100** while maintaining a minimal **drift score of 0.0043**.
- These results demonstrate the effectiveness of continuous Responsible AI monitoring and governance.
- The framework integrates explainability, fairness assessment, stability monitoring, drift detection, governance scoring and sustainability analysis into a unified Responsible AI governance solution.



Trustworthy AI



Fair & Unbiased



Transparent



Accountable

